# Appendix F - Math and Derivative Helpers

Numeric and string helpers BASIC provides at the prompt. Each one is documented in full in Chapter 2; this appendix is the single-page reference.

## F.1 Numeric functions

| Function | Domain | Result |
|---|---|---|
| `ABS(x)` | any number | Absolute value $\lvert x \rvert$. |
| `SGN(x)` | any number | `-1` if `x < 0`, `0` if `x = 0`, `1` if `x > 0`. |
| `INT(x)` | any number | Integer part of `x`, truncated toward zero. |
| `SQR(x)` | `x >= 0` | Square root. |
| `EXP(x)` | any number | `e^x`. |
| `LOG(x)` | `x > 0` | Natural logarithm. |
| `SIN(x)` | radians | Sine. |
| `COS(x)` | radians | Cosine. |
| `TAN(x)` | radians | Tangent. |
| `ATN(x)` | any number | Arctangent, result in `(-pi/2, pi/2)`. |
| `RND(x)` | sign-sensitive | `x > 0`: next pseudo-random in `[0,1)`. `x = 0`: repeat last value. `x < 0`: reseed with `INT(x)` and emit first draw. |
| `FRE(x)` | argument ignored | Bytes of free BASIC program / variable storage. |
| `POS(x)` | argument ignored | Current cursor column on the terminal. |
| `USR(x)` | address | Call an IE64 user machine-code routine; see Chapter 2. |
| `MEMALLOC(size[,align])` | byte count, optional alignment | Allocate a public low32 buffer and return its exact address. |
| `PEEK(a)` | address | Byte-width read; alias for `PEEK8(a)`. |
| `PEEK8(a)` | address | Byte-width read. |
| `PEEK16(a)` | address | `16`-bit aligned read. |
| `PEEK32(a)` | address | `32`-bit aligned read. |
| `PEEK64(a)` | address | `64`-bit aligned read; returns an exact integer qword. |

## F.2 String functions

| Function | Result |
|---|---|
| `LEN(s$)` | Number of bytes in `s$`. |

| Function | Result |
|---|---|
| `VAL(s$)` | Numeric value parsed from the start of `s$`. |
| `ASC(s$)` | ASCII code of the first byte of `s$`. |
| `CHR$(n)` | One-byte string with ASCII code `n`. |
| `STR$(n)` | Decimal text form of `n`, with a leading space for non-negative numbers. |
| `LEFT$(s$,n)` | First `n` bytes of `s$`. |
| `RIGHT$(s$,n)` | Last `n` bytes of `s$`. |
| `MID$(s$,p[,n])` | `n` bytes of `s$` starting at byte position `p` (1-based). With one argument, returns everything from `p` to the end. |
| `TAB(n)` | Move cursor to column `n`; only valid inside `PRINT`. |
| `HEX$(n)` | Uppercase hexadecimal text form of `n`. |
| `BIN$(n)` | Binary text form of `n`. |

## F.3 Derived identities

These are not separate functions: they are equivalences a program can rely on when composing the helpers above.

| Wanted | Compute as |
|---|---|
| natural log | `LOG(x)` |
| base-10 log | `LOG(x) / LOG(10)` |
| base-2 log | `LOG(x) / LOG(2)` |
| arcsine of `x` | `ATN(x / SQR(1 - x*x))` |
| arccosine of `x` | `1.5707963 - ATN(x / SQR(1 - x*x))` |
| truncated remainder | `x - INT(x / m) * m` |
| round to nearest | `INT(x + 0.5)` for `x >= 0`; `INT(x - 0.5)` for `x < 0` |
| fractional part | `x - INT(x)` |
| min(a,b) | `(a + b - ABS(a - b)) / 2` |
| max(a,b) | `(a + b + ABS(a - b)) / 2` |
| pi | `4 * ATN(1)` |
| e | `EXP(1)` |

## F.4 Range and precision

Numeric functions return BASIC double-precision values, except for helpers such as `PEEK64` and `MEMALLOC` that deliberately return exact integer payloads for hardware work. The trigonometric helpers accept any radian argument; very large arguments lose precision in the usual way during range reduction. Programs that need accuracy across many octaves of input should fold the argument into `[-pi, pi]` themselves.

`RND` is a 32-bit linear-congruential generator. Its sequence is deterministic and reproducible after a `RND(-seed)` reseed; the first draw immediately after the reseed depends on `seed` and the generator's fixed multiplier.